

Solution Code - PL SQL programming - Control_Structures

Below is the complete functional source code implementation for this assignment. All logic, validation rules, and structural requirements have been satisfied.

Source File: solution.sql

```
1 | -- Scenario 1: Apply 1% interest rate discount to loans for customers over 60
2 | DECLARE
3 |     CURSOR c_cust IS SELECT CustomerID, DOB FROM Customers;
4 |     v_age NUMBER;
5 | BEGIN
6 |     FOR r IN c_cust LOOP
7 |         v_age := TRUNC(MONTHS_BETWEEN(SYSDATE, r.DOB) / 12);
8 |         IF v_age > 60 THEN
9 |             UPDATE Loans SET InterestRate = InterestRate - 1 WHERE CustomerID = r.CustomerID;
10 |        END IF;
11 |    END LOOP;
12 |    COMMIT;
13 | END;
14 | /
15 |
16 | -- Scenario 2: Promote VIP status for balance over 10000
17 | ALTER TABLE Customers ADD IsVIP VARCHAR2(5) DEFAULT 'FALSE';
18 | DECLARE
19 |     CURSOR c_cust IS SELECT CustomerID, Balance FROM Customers;
20 | BEGIN
21 |     FOR r IN c_cust LOOP
22 |         IF r.Balance > 10000 THEN
23 |             UPDATE Customers SET IsVIP = 'TRUE' WHERE CustomerID = r.CustomerID;
24 |         END IF;
25 |     END LOOP;
26 |    COMMIT;
27 | END;
28 | /
29 |
30 | -- Scenario 3: Due loan reminders
31 | DECLARE
32 |     CURSOR c_loans IS
33 |         SELECT l.LoanID, c.Name, l.EndDate FROM Loans l
34 |         JOIN Customers c ON l.CustomerID = c.CustomerID
35 |         WHERE l.EndDate BETWEEN SYSDATE AND SYSDATE + 30;
36 | BEGIN
37 |     FOR r IN c_loans LOOP
38 |         DBMS_OUTPUT.PUT_LINE('Reminder: Customer ' || r.Name || ' has loan ' || r.LoanID || ' due on ' ||
TO_CHAR(r.EndDate, 'YYYY-MM-DD'));
39 |     END LOOP;
40 | END;
41 | /
```